

Final Project Report: Brain Tumor Classification from MRI Scans

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Abstract

We investigate the classification of brain MRI images into three tumor categories: glioma, meningioma, and pituitary tumor. Our approach employs three different deep learning models: a custom 3D Convolutional Neural Network (CNN) built from scratch to leverage spatial volume, and two transfer learning models using pretrained ResNet18 and InceptionV3 architectures. Using a publicly available Brain Tumor Classification dataset, we preprocess the images into uniform formats and apply data augmentations to improve generalization. Our ResNet18 model achieved moderate success, while our 3D CNN and InceptionV3 models exhibit superior performance with accuracies of up to 96.6%, indicating the potential of models that leverage volumetric information when compared to deep convolutional networks. We analyze the effectiveness of each model, perform ablation studies on augmentation and slicing strategies, and discuss generalization, overfitting, and model interpretability.

1. Introduction

Brain tumors are abnormal growths in the brain that can significantly affect a person's health, cognitive abilities, and quality of life. There are several types of brain tumors, each with varying severity and required treatments. Among them, gliomas, meningiomas, and pituitary tumors are three of the most commonly diagnosed. [6] Accurate identification and classification of these tumors through MRI scans are crucial for early diagnosis, treatment planning, and prognosis. However, the manual classification of MRI images by radiologists can be time-consuming and prone to human error. This has led to increasing interest in using deep learning techniques to automate the classification process [7].

In this project, we aimed to develop a deep learning-based system capable of automatically classifying brain MRI images into one of the three tumor types: glioma, meningioma, or pituitary tumor. Our specific goal was to compare the performance of three different model archi-

tectures: (1) a custom 3D Convolutional Neural Network (CNN) built from scratch, (2) a transfer learning model based on ResNet18, and (3) a transfer learning model using InceptionV3. We evaluated these models on a publicly available dataset and analyzed their performance in terms of classification accuracy, generalization ability, and training efficiency.

This problem is significant because early and accurate classification of brain tumors can greatly improve patient outcomes by enabling timely and appropriate medical interventions. Automating this task can also reduce the workload of medical professionals and standardize diagnostic accuracy across institutions. [3] If successful, our system could potentially be used in clinical settings to assist radiologists in the diagnostic process, especially in regions with limited access to expert healthcare providers. More broadly, this work demonstrates the application of deep learning to critical real-world medical imaging problems, contributing to the growing field of AI-assisted diagnostics.

2. Related Works

Deep learning has emerged as a powerful tool in medical image analysis, with numerous studies demonstrating its potential in identifying abnormalities such as tumors in MRI scans. Our work builds upon a growing body of literature that leverages convolutional neural networks (CNNs) and transfer learning for medical image classification.

Several studies have explored object detection and segmentation of tumors within MRI scans. For instance, Abdusalomov et al. [1] utilized YOLOv7, an advanced object detection model, enhanced with Convolutional Block Attention Modules (CBAMs) to highlight relevant features within brain MRI images. Their work achieved an accuracy exceeding 99.5%, suggesting that attention-based models, especially those incorporating transformer-like architectures, can significantly improve performance. However, their focus was on segmentation and detection, whereas our objective is multi-class classification—identifying whether a scan corresponds to a glioma, meningioma, or pituitary tumor. Unlike their approach, we exclusively use convolutional architectures without additional attention modules.

Other research efforts have focused on dataset expansion and mining to increase the diversity and size of MRI scan collections [2]. While such approaches can improve generalization by exposing models to a wider variety of samples, they often introduce inconsistencies due to differing preprocessing methods. In contrast, we utilize a standardized and curated dataset obtained from a public online repository, which ensures uniform preprocessing and mitigates potential bias introduced by non-standardized imaging protocols.

Horasan and Güneş [4] demonstrated the effectiveness of ensemble learning in the context of brain tumor classification. Their approach combined predictions from three distinct CNN architectures, ultimately improving performance beyond 90% accuracy. This highlights the value of model diversity and ensemble methods in reducing classification variance. While our project does not currently implement ensemble techniques, their work opens a promising path for future improvements.

Additionally, prior research has leveraged transfer learning to adapt pretrained models—originally trained on large-scale datasets like ImageNet—to medical imaging tasks. This technique has proven effective when dealing with limited data [8]. Our use of ResNet18 and InceptionV3 is motivated by this precedent. By fine-tuning these architectures on our brain MRI dataset, we aim to harness the representational power of pretrained convolutional features while adapting them to our specific classification task.

In summary, while our project draws inspiration from established work in segmentation, detection, and ensemble learning, it diverges by focusing solely on classification using both a custom 3D CNN and transfer learning models. Our goal is to evaluate how well standard convolutional techniques, when applied individually, can distinguish between three major types of brain tumors in MRI scans.

3. Method / Approach

To address the classification of brain MRI scans into one of three tumor types—glioma, meningioma, and pituitary tumor—we implemented a structured, multi-model approach. Our methodology involves preprocessing, model selection, training, evaluation, and validation.

3.1. Dataset, Preprocessing, and Output

The dataset we used is the “Brain Tumor Classification MRI Images” dataset hosted on Kaggle by JarvisGroot [5]. It includes grayscale brain MRI images categorized into three tumor types: glioma, meningioma, and pituitary. The dataset is stored in pickle files, so it required a preprocessing algorithm to turn the data into an easily accessible file-based structure for training, testing, and validation.

For the 3D CNN model, we grouped adjacent slices into fixed-depth tensors, allowing us to leverage the spatial correlations between neighboring slices. This step required

consistent reordering and zero-padding of image stacks to create uniform tensor inputs. We performed data augmentation through random rotations and horizontal flips to improve generalization. For 2D models, we extracted representative slices from the center of the brain scan, which often highlight the tumor region most clearly.

For model output, the neural networks predicted a probability distribution over the three tumor types: Pituitary, Glioma, and Meningioma. The predicted class was naturally the class label associated with the highest probability.

3.2. Model Architectures

We explore three models to capture a comprehensive performance landscape:

3.2.1. Custom 3D CNN (Baseline)

Our baseline model is a 3D Convolutional Neural Network, chosen due to its capacity to capture volumetric features in MRI scans, which are critical for understanding the full structure of tumors. This network was designed from scratch with multiple 3D convolutional layers followed by 3D max pooling layers, batch normalization, and ReLU activation. The final feature maps are passed to a series of fully connected layers and a softmax output layer for classification. We added dropout layers to mitigate overfitting.

The 3D CNN was implemented in PyTorch. During training, we used the Adam optimizer and categorical cross-entropy loss. We tuned hyperparameters such as learning rate, batch size, and the number of epochs training lasted for.

3.2.2. ResNet18 (2D Transfer Learning)

We applied a pretrained ResNet18 model on 2D slices of the MRI scans. ResNet18 was chosen for its robustness to vanishing gradients via residual connections, which allow deep architectures to train more effectively. Most of the convolutional base of the model was frozen, with a random subset of the base being unfrozen to reach 2 million parameters; in addition to the unfrozen base layers, the final fully connected layers were fine-tuned for tumor classification.

This approach leverages feature representations learned from the ImageNet dataset, adapting them to medical imaging through domain-specific training. We applied two fully connected layers with dropout to tailor the model to our three-class problem.

3.2.3. InceptionV3 (2D Transfer Learning)

InceptionV3 was selected for its efficient computation and capability to process multi-scale features through parallel convolutional paths. Like with ResNet18, we froze most of the early layers and fine-tuned the top layers. This model is expected to be computationally efficient and perform well even with relatively fewer training samples.

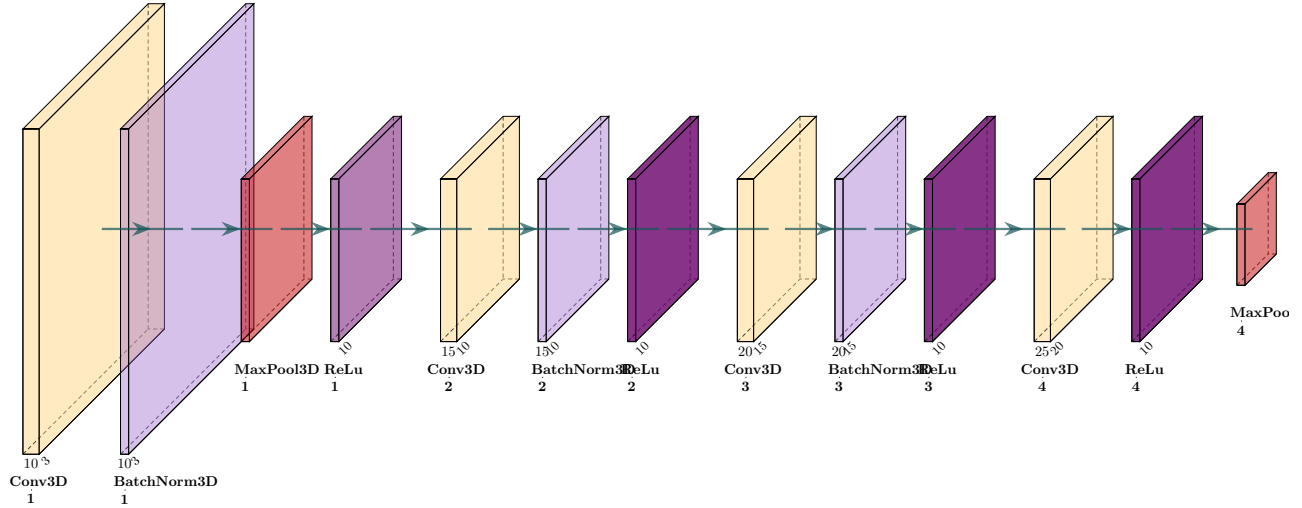


Figure 1. 3D CNN Architecture

3.3. Loss Function and Appropriateness

The categorical cross-entropy loss function was used for this classification task. This function is commonly employed in multi-class classification problems as it measures the performance of the model by comparing the predicted class probabilities with the true class labels. Given the categorical nature of the task (three tumor classes), this loss function was appropriate because it directly penalizes incorrect predictions in a way that supports optimization during training.

3.4. Evaluation Strategy

To evaluate model performance, we use accuracy and F1-score, with an emphasis on F1-score due to the potential for class imbalance. We further assess performance using confusion matrices, which allow for detailed insights into true positives, false positives, true negatives, and false negatives for each tumor type.

We split the dataset into training 80% and validation 20% sets. Additionally, we conducted ablation studies to evaluate the effect of various preprocessing techniques (e.g., slice count, normalization strategy), architectural changes (e.g., layer depth, dropout), and training configurations (e.g., learning rate).

3.5. Justification of Design Choices

The 3D CNN was designed to capture spatial dependencies within the MRI slices by using 3D convolutions, which can process 3D data and retain context across different slices. This architecture was expected to handle volumetric data in a way that would preserve spatial coherence between slices. On the other hand, the ResNet18 and In-

ceptionV3 models, based on pre-trained architectures, were used for extracting features from individual 2D slices by leveraging transfer learning. These models are highly efficient in extracting complex features from medical images, and their pre-trained weights helped in mitigating the challenge of limited data for training.

We considered using transformer-based architectures such as Vision Transformers (ViT) and hybrid models like Swin Transformers, which are becoming increasingly popular in medical imaging. However, given our limited computational resources and the focus on convolutional models earlier in the semester, we prioritized CNN-based methods for feasibility and interpretability.

3.6. Learned Parameters and Model Components

Our models were built primarily around convolutional layers, which contained the learned parameters. The convolutional layers of the CNN were responsible for feature extraction, while fully connected layers at the end of the networks were used for classification. Additionally, both ResNet18 and InceptionV3 architectures had learned parameters in their convolutional layers, which were fine-tuned for the task of classifying brain tumors.

In contrast, post-processing layers, such as the softmax layer for classification, did not have learned parameters. These layers simply performed a mathematical transformation of the model's outputs into probabilities representing the likelihood of each class.

3.7. Tools and Resources

We used PyTorch for all model implementations. Pre-trained weights for ResNet18 and InceptionV3 were sourced from torchvision.models. Training and evaluation

were conducted in Google Colab and with Jimmy’s beast of a GPU. The dataset used is available on Kaggle [5] and can be found by following any of the citations near its mentioning.

4. Data

The dataset used in this project is publicly available on Kaggle, titled ”Brain Tumor Classification (MRI)” by user JarvisGroot [5]. This dataset contains 2,450 MRI images, each labeled as either Glioma, Meningioma, or Pituitary tumor. The images are stored in PNG format and are organized into subdirectories corresponding to each tumor type.

Each image is originally 512x512 pixels and, while grayscale in nature, is loaded as a 3-channel image to accommodate the input requirements of convolutional neural networks such as ResNet18 and InceptionV3. The dataset does not explicitly provide patient metadata or volumetric scans; this means that combining slices into 3D datapoints requires an implicit assumption that every six images in a row belong to the same patient. This implicit assumption is likely not that harsh of an assumption based on our group’s visual analysis.

4.1. Source and Collection

The dataset was published on Kaggle [5], and while the precise origin of the images is not disclosed, it is implied they come from clinical imaging sources. The dataset is anonymized and manually labeled. No additional demographic or acquisition details are available.

4.2. Preprocessing

To prepare the images for training and evaluation, several preprocessing steps were applied. First, all images were resized to 224x224 pixels for ResNet and 299x299 pixels for InceptionV3 to be compatible with pretrained model architectures. Next, image intensities were normalized using the mean and standard deviation values from the ImageNet dataset. Augmentation techniques, such as random rotation, reflections, and Color Jitter were applied to increase data variance and help prevent overfitting.

4.3. Data Splitting

The dataset was divided using an 80-20 split for training and validation purposes. Stratified sampling was used to preserve the class distribution in both sets. A separate portion of the validation set was later used as a test set for final evaluation of model performance.

4.4. Biases and Limitations

There are a few notable limitations in the dataset. First, the lack of explicitly defined 3D MRI volumes limits the

ability to develop or evaluate fully volumetric models without making moderate assumptions about the dataset’s organization. Second, there is some degree of class imbalance, with Pituitary tumor images appearing more frequently than the others.

Despite these constraints, the dataset is suitable for evaluating CNN-based tumor classification models. Its quality and organization support effective experimentation with both custom 3D CNNs and 2D transfer learning pipelines.

5. Experimentation and Results

This section presents the experimental procedures, metrics, and outcomes that demonstrate the performance of our proposed models for the brain tumor classification task. Our goal was to evaluate and compare three deep learning architectures (3D CNN, ResNet18, and InceptionV3) using MRI data, and determine which approach best generalizes to unseen data.

5.1. Experimental Setup

5.1.1. Evaluation Metrics

We utilized two standard classification metrics to evaluate model performance: accuracy and F1-score. Accuracy was the primary metric, providing an overall sense of model correctness. However, F1-score was used to assess the model’s ability to correctly distinguish among the three tumor types, which is particularly critical for imbalanced or subtle medical imaging tasks.

5.1.2. Baselines

Our baseline was a custom 3D CNN built from scratch, designed to process a stack of grayscale MRI slices. The results from this model served as a benchmark to evaluate improvements from transfer learning approaches using pretrained 2D architectures.

5.1.3. Implementation Details

All models were implemented using PyTorch. The dataset was split into training and validation subsets using an 80:20 split, maintaining class balance in both subsets.

Training was conducted with the Adam optimizer, a batch size of 32, a starting learning rate of 1×10^{-4} , and a weight decay of 1×10^{-8} . The choice of optimizer, Adam, was made due to its adaptive learning rate properties, which help it converge more quickly compared to traditional stochastic gradient descent (SGD). Adam also tends to perform well in practice, especially with noisy or sparse data like medical images. Each model was trained for 20 epochs. The training and evaluation were performed on a GeForce RTX 4070, with 12GB of VRAM.

Some models needed finer adjustments, specifically InceptionV3. By default, the full InceptionV3 model has approximately 27 million parameters, which when compared

to its contesting models is magnitudes higher. The comparison between models would therefore not be fair in terms of relating architectural design or computational power. To combat this, the model first had all but a select few layers frozen from training, which lowered the parameter count to be relative to its other models.

The number of parameters for each model is as follows:

- 3D CNN: 1,127,063 parameters
- ResNet18: 2,228,163 parameters
- InceptionV3: 2,447,598 parameters

5.2. Quantitative Results

The performance of each model is summarized in Table 1.

Table 1. Quantitative Comparison of Classification Accuracy

Model	Training Accuracy	Validation Accuracy
3D CNN	92.33%	90.38%
ResNet18	86.29%	85.15%
InceptionV3	95.18%	95.60%

Table 2. Quantitative Comparison of F1-Score

Model	F1-Score
3D CNN	89.95%
ResNet18	83.07%
InceptionV3	94.93%

5.3. Qualitative Results

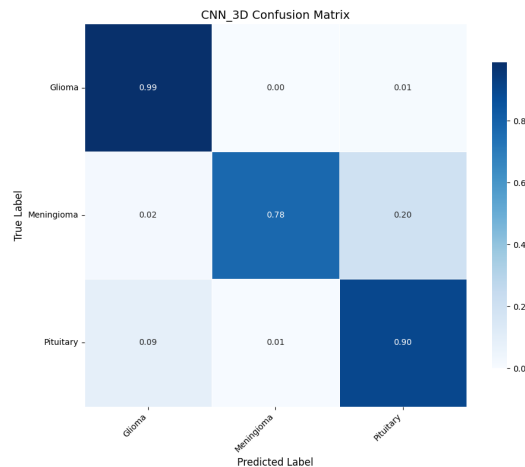


Figure 2. Confusion matrix for 3D CNN model.

Confusion matrices from each model are illustrated in Figures 2, 3, 4. The InceptionV3 displays very high consistency in predicting tumor types correctly, while the

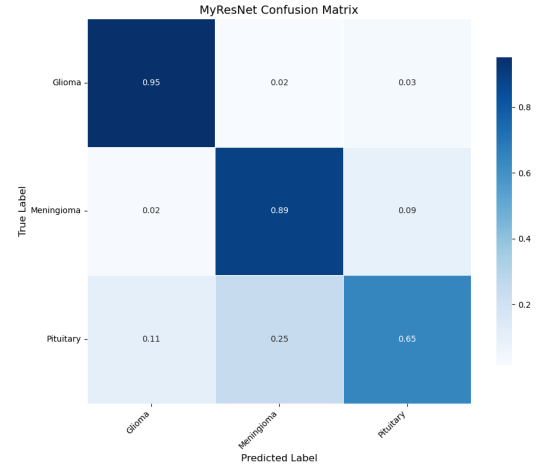


Figure 3. Confusion matrix for ResNet18 model.

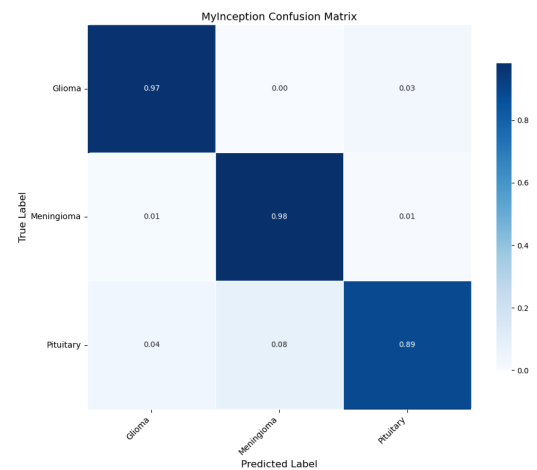


Figure 4. Confusion matrix for InceptionV3 model.

ResNet model occasionally misclassifies Pituitary Tumors as Meningioma. These results show that both our specially designed 3D convolutional network and the transfer learning from InceptionV3 were able to be well applied to this domain. ResNet seems to do worse than the other two models because of its lower depth when compared to InceptionV3 made it a worse overall feature extractor. We did observe that when leaving all parameters of ResNet18 as learnable parameters, a validation accuracy greater than 96% was achievable, but this would require ~ 8 million parameters, making a comparison with the other models unfair.

5.4. Hyperparameter Tuning and Ablation

For the 3D CNN, multiple configurations were explored, including variations in convolutional layer depths, filter sizes, and pooling strategies. The selected architecture

achieved the best validation accuracy when using 4 convolutional layers followed by batch normalization, max pooling, and ReLU activations.

Several hyperparameters were fine-tuned during training, including the learning rate, batch size, and the number of epochs. We used a learning rate of 1×10^{-4} with weight decay of 1×10^{-8} to prevent overfitting and ensure smooth convergence. The batch size was set to 32 to balance computational efficiency and memory constraints, and the models were trained for 20 epochs.

5.5. Analysis and Discussion

Among the three models, the InceptionV3 outperformed the other architectures in training and validation accuracy by a large margin, while the ResNet18 model performed particularly bad in comparison to the other two models.

We observed that the 3D CNN model, while having only around 1.13 M parameters, reaches a solid validation accuracy (90.38%), indicating it generalized well to unseen data. However, the ResNet18 model demonstrated slightly lower performance in validation, which suggested a potential overfitting issue. The InceptionV3 model, on the other hand, showed equally high accuracy on both the training (95.18%) and validation (95.60%) sets, indicating good generalization, likely due to its ability to leverage pre-trained weights from a larger dataset.

The custom 3D CNN achieved solid performance given its relatively small parameter count, indicating that even a straightforward volumetric CNN can be effective when trained on stacked MRI slices. While ResNet18 and InceptionV3 leverage large pretrained weights, the gap with our lightweight 3D CNN is surprisingly small, only about 4 % behind InceptionV3, highlighting that a well-tuned volumetric model, when paired with aggressive augmentations and regularization, can rival standard 2D approaches.

It is to be expected that the pretrained models would perform the best in their categories, as they allow for deeper learning capabilities with advanced architectural features such as residual connections or auxiliary classifiers respectively. The ResNet18 model underperformed relative to both 3D CNN and InceptionV3. One plausible reason is that ResNet18, though pre-trained on ImageNet, may not adequately capture subtle MRI features due to the shallower depth and reduced representational capacity. Furthermore, it showed signs of overfitting, as validation accuracy plateaued early during training.

Despite the already strong results from InceptionV3, there is a large room for improvement. We observed accuracies greater than 99% when all layers were learned parameters, requiring over 25 million parameters limiting its utility based on available hardware.

5.6. Limitations

One limitation of our approach is that the MRI slices in the dataset were not explicitly connected to individual patients. As a result, we had to make an implicit assumption that every six consecutive images in a row belonged to the same patient. This assumption may introduce some uncertainty and could affect the model's ability to capture patient-specific patterns.

6. Conclusion

In this project, we developed a multi-model approach for classifying MRI scans into three types of brain tumors: Pituitary, Glioma, and Meningioma. Our models, including a custom 3D CNN, ResNet18, and InceptionV3, demonstrated promising results. The 3D CNN achieved a validation accuracy of 90.38%, while the ResNet18 and InceptionV3 models performed at 85.15% and 95.60%, respectively. These results highlight the effectiveness of transfer learning with InceptionV3, although the custom 3D CNN provided a strong baseline for future improvements.

Through our experiments, we learned that while the models were able to classify MRI scans effectively, the lack of patient-specific data tied to each MRI slice presented a limitation. This assumption may have affected model performance in capturing patient-specific patterns and variability. We also found that using different architectures like InceptionV3 resulted in a significant increase in validation accuracy, indicating that pre-trained models can help address complex medical image classification problems.

Future work will focus on refining dataset preprocessing by associating MRI slices with individual patients, incorporating ensemble methods for improved accuracy, and exploring advanced architectures such as Transformer-based models and U-Net for segmentation tasks.

This work could be extended to other medical imaging tasks, such as diagnosing other types of tumors or identifying abnormalities in different modalities like CT scans. The techniques we used may also be applicable to other classification problems in healthcare, where large amounts of labeled image data are available but not necessarily tied to specific patient metadata.

7. Team Contributions

Student Name	Contributed Aspects	Details
William Akins	Trainer-Implementation, ResNet18-Implementation, Report	Created a trainer class for model experimentation, working for both 3D and 2D models; fine-tuned ResNet18 with a couple final fully connected classifier layers; wrote the major text sections of the report.
Jimmy Vu	Data collection & processing, InceptionV3 Implementation, Experiment Analysis	Created an ImageLoader class (subclass of Torch Dataset class) to house data; created functions to transform data; fine-tuned InceptionV3 with a final fully connected layer; Ran and experimented with all the models.
Chen Zhang	Visualizations Implementation, 3D CNN Implementation, Report	Created a Visualizer class to create loss and accuracy graphs as well as confusion matrices; implemented 3D CNN model; added final edits on report as well as adding images to report.

Table 3. Contributions of team members.

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